**Due:** Friday, April 3, 2026, 11:59 AM

## Assignment 3: Build 3 - HITL + Tool Router Agent

### This is the first team assignment.

In this assignment, you will add an LLM router to the HITL code to generate data analysis python code, human code approval, and agent tool routing. The class demo file is in the Github repo: *builds/build3_hitl_router_agent.py*. You can edit this code for your own project

The application you build can be a part of your bigger course project, so you can build something that is tailored toward the goals of your project. However, **the AI assistant should be built using LangChain, OpenAI, and Langfuse**. At the very minimum, this agent should use the data set you have chosen for your team project.

Your agent should be designed to help automate data analysis decisions and enhance a component of your project pipeline, such as coding, data cleaning, statistical analysis, visualization, model interpretation, or report generation. The goal is to use this assignment to create something that moves your project along. The workflow should be capable of guiding users through other data-related tasks and analyzing data.

**The core requirements of your agent are that it:**

1. Is implemented using LangChain with an OpenAI LLM
2. Accepts user input in the form of a question, task, or instruction related to a dataset or data analysis goal and returns code and data analysis results and summaries.
3. Generates python code upon request, seeks human code approval, and has an LLM to decide whether to use a tool from an allow-list or to fall back on code generation if there is no appropriate tool.
4. Executes without error and returns output to the user.
5. Provides a natural language explanation/interpretation of the output.
6. Is evaluated using Langfuse tracing.

**You should build your application in your ai-data-analysis-agent virtual environment**.

**The deliverable for this assignment is a GitHub repository for your project that is shared with the instructors and TAs.** The repository should include:

1. A README.md file that includes purpose of the application, instructions for using the app, and cautions for using the app.

2. A requirements.txt file with your project dependencies.

3. A **build3_hitl_tool_router_agent.py** file with the code that develops and executes the app.

4. The data set used to develop the application.

5. A document that contains a tracing and error log from Langfuse and a summary of the agent's performance including any issues you encountered and how the issues were resolved.

Add submission

### Submission status

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|---|---|
| **Attempt number** | This is attempt 1. |
| **Submission status** | No submissions have been made yet ? |

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| **Grading status** | Not graded |
| **Time remaining** | 2 days 2 hours remaining |
| **Last modified** | - |
| **Submission comments** | Comments (0) |